ANSYS Drone Analysis

I. Introduction

Drones have witnessed a large amount of development in commercial and recreational uses in the last decade. The drones with four arms, also known as quadcopters, are some of the most common types of drones in use today. They have four motor-propelled blades that are symmetrically aligned around the center hub. Because of their simple design and how easy it is to control, these quadcopter-based drones are the most popular type of drone. They are used for things like videography, photography, exploration, and even delivering packages.

The overall design of a drone is extremely important because of its potential uses. On one hand, the frame should be lightweight to maximize efficiency; it's important to remember that drones do not have particularly large motors to power themselves. Not only that, but they need to be aerodynamic for the same reason of efficiency. However, these drones still need to be strong and durable so they can serve their variety of purposes. What this means is that the frame of the drone needs to have minimal mass while also being able to sustain different types of forces such as wind, and impacts from landing. There are four major parts in the construction of this particular drone, the body hub, four arms, blades, and the landing gear system.

Important loads on the drone include the forces from the thrust produced by the drone from its blades, the load due to gravity, and air resistance. This is one of if not the most important parts of an ANSYS analysis because it shows what may happen in a real-world scenario. It is important for the drone design such that the safety factor with respect to yielding remains greater than the minimum acceptable value for all possible loading scenarios.

This paper aims at carrying out an FEA of a drone using ANSYS Mechanical software. It involves modeling the geometry in SpaceClaim, mesh generation, loading and boundary conditions, mesh convergence checking, and failure analysis.

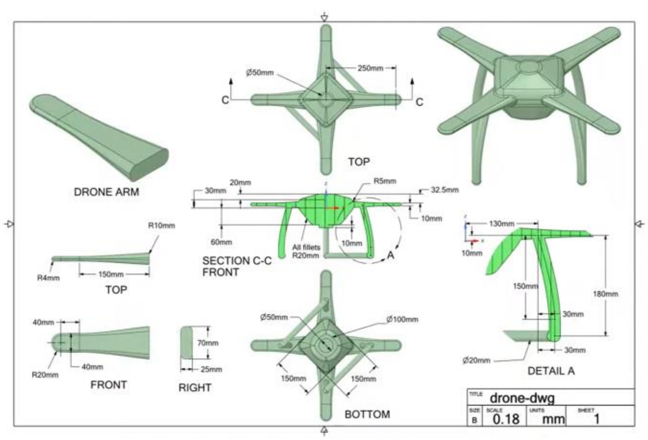

Figure 1 – Dimensions of Drone Body [1]

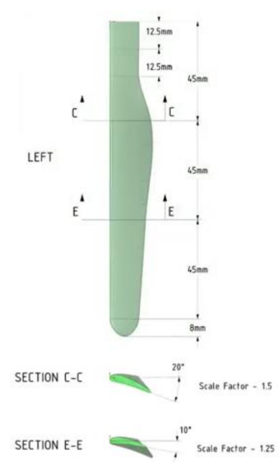

Figure 2 – Dimensions of Drone Wing [2]

II. Geometry Model

The geometry of the drone was made using SpaceClaim, starting from basic individual parts like the propellor and then adding onto the drone body until the full geometry was completed.

The drone blade, also called the propellor subassembly, was the first part made. The blade profile was first sketched in a shape similar to the cross-section of an airplane wing. It was duplicated into 5 separate planes, some of which were oriented and sized differently to fit the shape of the blade. Then the overall shape of the blade was formed by connecting these segments into one fluid piece. Because of the way the planes were formed it gave a long narrow shape along with a slight twist throughout the blade. This twist is what allows the blade to generate lift as it spins. Once one blade was finished, it was duplicated and rotated 180 degrees around the hub to create a 2-blade propeller assembly.

It is interesting to note that as mentioned earlier, the original sketch was like the cross section of an airplane wing. The way the drone and the airplane generate lift is a little different though. The airplane forces air over the wings faster than the air under the wings so there is a low-pressure zone above the wings, the drone, for each propeller assembly, spins each blade and because of the twist in the blades it forces air downwards creating lift. The geometry of the propellor assembly can be seen below.

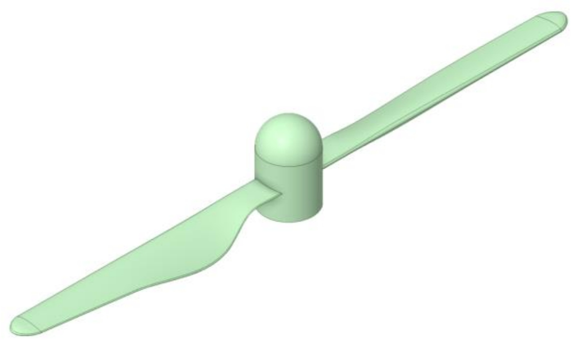

Figure 3 – Isometric View of Geometry of Propellor Assembly

The next step was building the central body. If this were a real drone, then things like electronics and batteries would be here. It is a unique shape to describe, the first step in creating it was creating a sketch of a square, then providing sketches of circles above and below which were all together conjoined into a box-like shape with angled sides, giving it the slightly hexagonal appearance. An important part of this step was filets; this gives the drone a more seamless and fluid look.

Next, the arms were created; a single arm started off referenced from the main body; however, immediately the main body of the drone was hidden to allow the arm to be designed without interference from the main body. The overall shape was created using similar techniques as mentioned before. Similarly to the drone blades, the arms were duplicated and rotated around the body 360 degrees to create 4 arms.

Finally, the landing gear was constructed beneath the central body. This was the simplest part to create, using the splice tool three separate lines were drawn on one side of the drone body. This can be seen in three separate segments of each landing leg. Then, while selecting all of them, the cylinder tool was used to provide volume. Again, it was duplicated to the other side of the drone's body.

With the rest of the geometry finished, the completed propeller assembly created earlier was imported into this geometry, using the assembly tool the propellers were properly aligned with the arm and once more duplicated and rotated around to each arm to give each of the four arms a propellor assembly.

The two views shown below capture the final product: a top-down view and a full isometric view of the assembled drone.

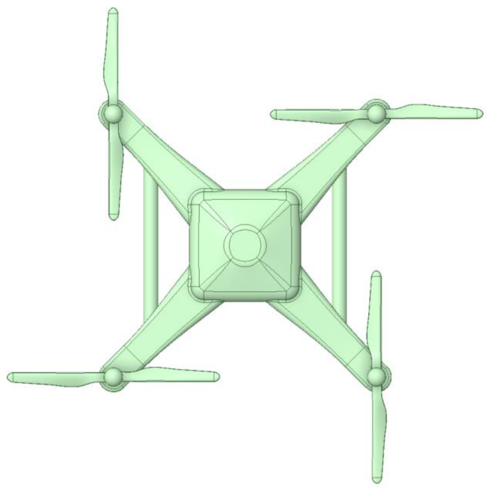

Figure 4 – Top View of Geometry of Total Drone Assembly

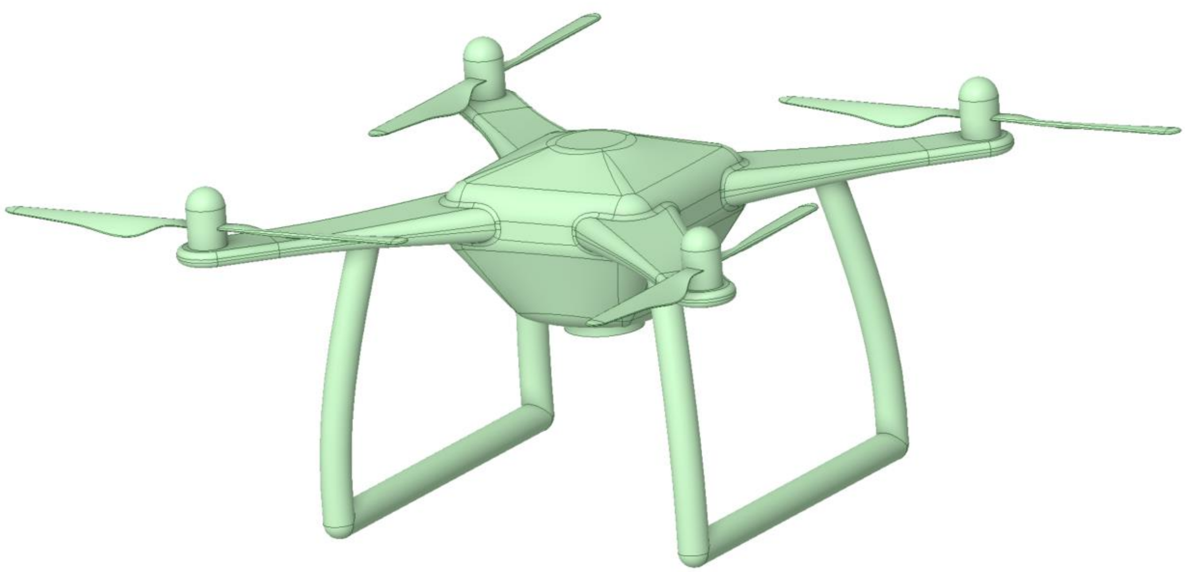

Figure 5– Isometric View of Geometry of Total Drone Assembly

III. Finite Element Mesh, Loads and Boundary Conditions

This part was done using ANSYS Mechanical; it is a straightforward process but extremely important. This is the groundwork that influences the analysis and if any steps are missing or not appropriate, then the analysis will either not work at all or not be accurate to what is trying to be solved.

The first step is to assign the material to each part of the geometry; the material used was an ABS high-impact plastic that would typically be found in drones. Next, the mesh had to be generated; this was done by first specifying an element size of 0.002 meters. The mesh was then generated and can be seen below. Something to note is how small the element size looks when looking at an isometric view of the full drone. This is beneficial for accuracy of the analysis but may be an issue visually down the road when the mesh is refined and the element size decreases.

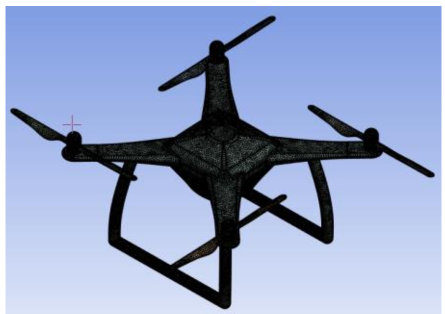

Figure 6 – Isometric View of Drone Mesh

After the mesh was generated, the next step was applying the forces to simulate a real environment for the drone. First, a gravity force was applied to the entirety of the drone, simulating earth's gravity. Next, an acceleration of 25 m/s was applied in an upward direction to simulate the flight of the drone. Next loading needs to be defined for each of the four arms and to do this a force of 5 newtons was applied in a downward direction for each arm. Finally, a force of 10 newtons is applied to the center hub to simulate the weight from the battery. The forces applied to the drone can be seen below.

Figure 7 – Top view of Loads Applied to Drone

The final step of this part of the project is assigning boundary conditions to the corners of the center hub of the drone. The first one is a fixed support; this simulates completely rigid motion where no moment or deformation is allowed. The second one is a displacement boundary condition; this one restricts only the x-component of displacement allowing y and z to be free. The final boundary condition is the second displacement which restricts the y-component allowing x and z to be free. There is technically a boundary condition at the fourth corner which can be considered another displacement boundary conditions, but no components are restricted. The boundary conditions can be seen below.

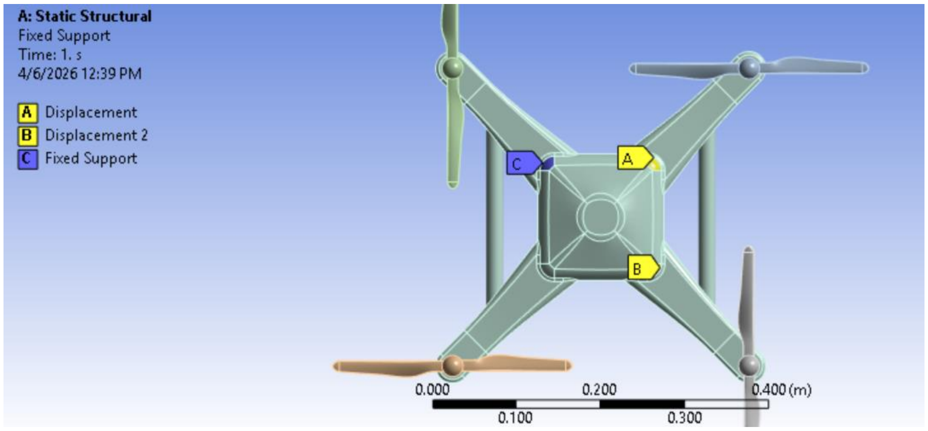

Figure 8 – Top View of Drone Boundary Conditions

IV. Static FE Analysis

The first step in a static finite element analysis is to assign whatever is desired to be analyzed; in this case it was total deformation as well as equivalent stress (von-misses). In the first picture, the deformation as well as the original shape is shown. It's important to know that the deformation is exaggerated because of the scaling that was applied; the actual physical displacements are much smaller than they appear. This was applied intentionally to show a clear difference in what is being deformed due to the applied loads. The most obvious deformation is in the blades; this makes sense given that they are the thinnest part of the structure. The deformations in the blades are seen bending upward due to the 5-newton force that was applied downward at each of the propeller locations. The central body and arms, being thicker and more supported by the boundary conditions, show much less visible deformation by comparison.

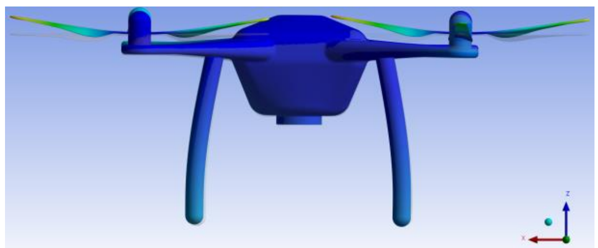

Figure 9 – Side View of Drone Deformation

This next image shows the results of the equivalent stress across the drone. By analyzing where the actual max stresses are it is seen to be around 1.5 MPa with the scale going to 3.46 Mpa; this happened where the blade meets the hub. This is what is expected because of what the blades do, since they experience a large amount of force compared to the small area where they connect to the hub. Mathematically this correlates because the more force applied in the smaller area, the larger the stress is going to be. Additionally, stresses appear at the corners of the central body where the boundary conditions were applied. This is also expected; fixed supports inhibit movement where they are located, which causes that location to experience more stress. The rest of the drone, including the arms, landing gear, and most of the body, remains at a lower level of stress.

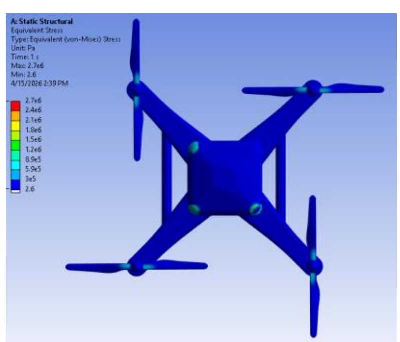

Figure 10 – Top View of Equivalent Stress

An important part of a finite element analysis is a force balance check, as seen below the forces balance which shows the validity of the analysis.

Force Reaction (Z) [N]	Force Reaction (Total) [N]
-26.151	26.599

Table 1 – Table showing Force Balance Check on Drone

To make sure of accurate results, a mesh refinement was performed. The element size went from 0.002 meters to 0.0015 meters. Because the refined mesh elements are so small, it is very hard to see the actual elements when looking at them from an isometric view; this is why a close-up view of the mesh as well as the equivalent stress is included to show the actual elements. After solving it again with the refined mesh, the equivalent stress was compared to those from the original mesh. The locations of the stresses were the same, with the noticeable stresses still located where the blade met the hub and at the boundary condition locations. By analyzing the results, the maximum observed stress is closer to 1.54 MPa with the scale going up to 3.46 MPa; this shows a higher level of analysis. The value to take from this analysis should come from the more refined mesh due to the elements being both smaller and more of them. Mesh refinement is important to make sure that the solution is accurate enough.

Figure 11 – Isometric View of Refined Drone Mesh

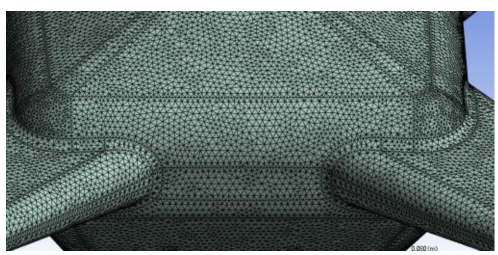

Figure 12 – Close up View of Refined Mesh

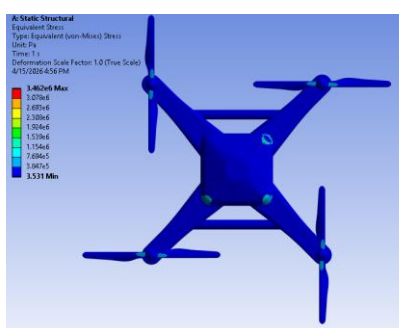

Figure 13 – Top View of Equivalent Stress of Refined Mesh

V. Failure Analysis

The goal of this section is to determine whether the drone frame is at risk of failing under the applied loads and the applied conditions. To do this, three results were found: total deformation, equivalent stress, and the safety factor. The safety factor is calculated by dividing the material's strength by the maximum equivalent stress at any given point in the model.

$$Factor\ of\ Safety\ (FoS) = \frac{Capacity\ (Material\ Strength)}{Demand\ (Allowed\ Stress)}$$

Equation 1 – Factor of Safety Equation [3]

A safety factor greater than one means the material is not expected to fail at that location, while a value below one would indicate that the stress has exceeded the yield strength and failure is predicted.

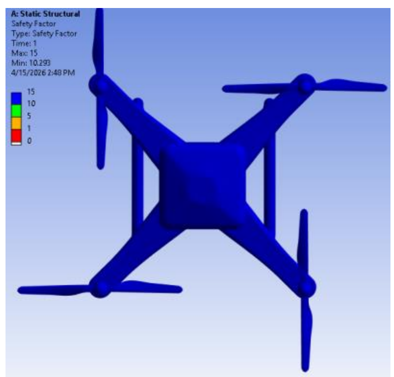

Figure 14 – Safety Factor of Drone for Failure Analysis

Looking at the results, the factor of safety shows the entirety of the drone's body is blue, which means a factor of safety around 15. This indicates that those parts are well protected against failure under the given loads. This tells us that no part of the drone is predicted to fail under the loads that were applied.

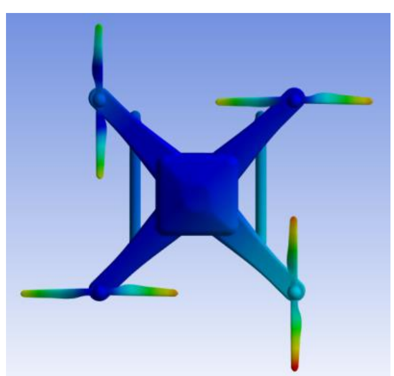

Figure 15 – Total Deformation for Failure Analysis

Looking at figure 15, we can see the results of the total deformation for failure analysis, again the deep blue represents low deformation, we can see higher deformation at the propellor assembly which again makes sense given the applied conditions and requirements of the drone. A similar result is seen below in figure 16 for the equivalent stress; we see higher levels of stress around the propellor component as well as where the boundary conditions were applied.

Figure 16 – Equivalent Stress for Failure Analysis

Overall, the failure analysis of the drone was good; the material (ABS high-impact plastic) is up to the task of handling the forces that the drone will experience. The large safety factor across the drone means that, if necessary, some material could be taken away to make the drone lighter, or the drone could be exposed to harsher loads and conditions. The only thing worth being concerned about could be aspects of the propellor assembly.

VI. Summary

This paper showed a complete finite element analysis of a drone using ANSYS Mechanical. The goal was to model the drone geometry, set up a realistic loading scenario, run a static structural analysis, verify the results through mesh refinement, and evaluate the structural integrity of the design through a failure analysis.

The geometry was built in SpaceClaim, starting with the propeller blades and then building the body, arms, and landing gear until a complete drone assembly was produced. The model was then brought into ANSYS Mechanical where it was assigned ABS high-impact plastic as the material, meshed with an element size of 0.002 meters, and loaded with a combination of Earth gravity, an upward acceleration of 25 m/s^2, a 5-newton downward force at each of the four propeller locations, and a 10-newton force at the center hub representing the battery weight. The applied boundary conditions were a fixed support, and two directional displacement constraints were applied to the corners of the central body to properly restrain the model.

The static analysis showed that the propeller blades had the greatest deformation and the highest stress levels, specifically at the point where each blade connects to the motor hub. The rest of the structure, including the arms, body, and landing gear, had low stress. Mesh refinement from 0.002 meters down to 0.001 meters confirmed that the results were consistent between the two meshes, verifying that the solution had converged and that the mesh was producing accurate results.

The failure analysis also showed that the drone design is satisfactory under the applied loads. The safety factor across the entire model ranged from approximately 10.22 at its lowest to 15 at its highest, with the minimum values located again where the propellor blade meets the hub. Since every part of the structure maintained a safety factor well above 1, no yielding is predicted under these loading conditions. The most important takeaway from this analysis is that while the overall design is safe, the propeller blades are consistently the most stressed and most deformed component and is the only part of the drone that would need to be worried about.

References

[1] Ansys Learning. "Solid Modeling of a Drone — Lesson 3." *YouTube*, 9 Feb. 2021, www.youtube.com/watch?v=4JhNpawwAD8. Accessed 25 May 2025.

[2] Ansys Learning. "Solid Modeling of a Drone Blade — Lesson 2." *YouTube*, 9 Feb. 2021, www.youtube.com/watch?v=aO1Hot3XxHs. Accessed 22 Apr. 2026.

[3] Cyprien. "FEA for All." *FEA for All*, 26 Dec. 2016, feaforall.com/calculate-safety-factor/.